Effect of density and magnetic field fluctuations on rf waves propagation

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1 Introduction

Low frequency fluctuations of the electron density or the magnetic field can strongly affect the propagation of rf waves in a magnetized plasma. This effect has been extensively investigated for the Lower Hybrid (LH) wave by solving a wave kinetic equation in the weak turbulence approximation [1, 2]. In this approach, the flow of rf energy is described by a usual ray-tracing between random scattering events where the initial wave vector $\mathbf{k}$ can be modified both in amplitude and direction with the constraint to remain solution of the local dispersion relation.

This physical mechanism is a good candidate for explaining a possible strong upshift of the power spectrum as the LH wave propagates in the plasma, even if the toroidal mode coupling for bridging the spectral gap is very weak. Though

primarily dedicated to the LH wave, the concept of wave scattering by low frequency density or magnetic field fluctuations can be easily extended to other types of rf waves provided $\widetilde{k}_{\parallel} \ll k_{\parallel}$ and $\widetilde{\omega} \ll \omega$. Here, $k_{\parallel} = \mathbf{k} \cdot \widehat{\mathbf{b}}$ and $\widetilde{k}_{\parallel} = \widetilde{\mathbf{k}} \cdot \widehat{\mathbf{b}}$ are the projections of the rf and fluctuation wave vectors respectively along the local direction of the magnetic field $\widehat{\mathbf{b}} = \mathbf{B}/\left\|\mathbf{B}\right\|$, while ω and $\widetilde{\omega}$ are the rf and fluctuation frequencies respectively. Consequently, the scattering process becomes a two dimensional problem in the plane perpendicular to the local magnetic field direction, a reasonable assumption regarding the large difference between the long wave length of the low frequency fluctuations of the plasma and the short length of the rf waves along $\widehat{\mathbf{b}}$.[1]

Calculations are organized in three steps. In the first section, the components of the scattered rf wave vector $\check{\mathbf{k}}$ are calculated, according to the curvilinear coordinate system chosen for describing the magnetic equilibrium in C3PO [5]. The wave kinetic equation corresponding to the wave-wave interaction is then derived from first principles in the second section for any kind of rf waves and plasma model. Finally, in the third section, the wave kinetic equation is solved using a Monte-Carlo method and the procedure for the numerical implementation in the code is explicited.

1.0.1 Calculation of the scattered wave vector

When the rf wave vector $\mathbf{k}$ is scattered by low density fluctuations, it may change both in module and direction, with the condition

$$\mathbf{k} \cdot \widehat{\mathbf{b}} = \check{\mathbf{k}} \cdot \widehat{\mathbf{b}} = k_{\parallel} \tag{1}$$

while $\check{\mathbf{k}}$ is still solution of the local dispersion relation. Consequently, the scattering process may be expressed in the simple form

$$\mathbf{k}_{\perp} \cdot \check{\mathbf{k}}_{\perp} = k_{\perp} \check{k}_{\perp} \cos\beta, \tag{2}$$

where $\check{k}_{\perp} = \left\|\check{\mathbf{k}}_{\perp}\right\| = \left\|\check{\mathbf{k}} \times \widehat{\mathbf{b}}\right\|$ may change by wave-wave interaction if several modes of propagation are solutions of the dispersion relation for the same $k_{\parallel}$, while β is the angle of the rotation, between the initial and scattered wave vector directions in the plane perpendicular to $\widehat{\mathbf{b}}$.[2] Unlike transitions are defined as those corresponding to $\check{k}_{\perp} \neq k_{\perp}$, while like ones to $\check{k}_{\perp} = k_{\perp}$.

Using the coordinate system defined in Ref. [5], the components $\left(\check{k}_{\rho}, \check{m}, \check{n}\right)$ of $\check{\mathbf{k}}$ are thus fully determined by the components (k_{ρ}, m, n) of the wave vector $\mathbf{k}$ prior the scattering event and the two parameters β and $\chi = \check{k}_{\perp}/k_{\perp}$.

[1] The parallel wave length $\lambda_{\parallel}$ is related to the parallel refractive index $N_{\parallel}$ of the rf wave by the simple relation $\lambda_{\parallel} = c/fN_{\parallel}$,where $f = \omega/2\pi$ is the rf wave frequency. Using typical values corresponding to rf systems used on the Tore Supra tokamak [3, 4], $f_{LH} = 3.7GHz$ and $N_{\parallel LH} \simeq 2$, $\lambda_{\parallel LH}$ is about $4cm$ for the LH wave, while it is $\lambda_{\parallel EC} \simeq 1.3cm$ for the Electron Cyclotron (EC) wave with $f_{EC} = 110GHz$ and $N_{\parallel EC} \simeq 0.2$. In both cases, the condition of validity of the model $\lambda_{\parallel} \ll \widetilde{\lambda}_{\parallel}$ where $\widetilde{\lambda}_{\parallel} \simeq qR_p$ is well fullfiled. since the safety factor $q > 1$, and $R_p \sim m$.

[2] In Ref. [1], the relation $\widehat{\mathbf{b}} \cdot \left(\mathbf{k}_{\perp} \times \check{\mathbf{k}}\right) = k_{\perp}\check{k}_{\perp} \sin\beta$ is also introduced. However, it is fully equivalent to (2) and does not provide any additional information.

The conservation (1) of the parallel component of the wave vector in the scattering process may be expressed in the form,

$$\sigma_I P \frac{\check{m}}{r} \cos\alpha + \sigma_B T \frac{\check{n}}{R} = \sigma_I P \frac{m}{r} \cos\alpha + \sigma_B T \frac{n}{R} \tag{3}$$

which gives the first equation which must be satisfied by components of $\check{\mathbf{k}}$ defined as

$$\check{\mathbf{k}} = \check{k}_\rho \left\|\nabla\rho\right\| \widehat{\rho} + \frac{\check{m}}{r}\widehat{\theta} + \frac{\check{n}}{R}\widehat{\phi} \tag{4}$$

according to [5]. Here, $\widehat{\mathbf{b}} = \sigma_B T\widehat{\phi} + \sigma_I P\hat{\mathbf{s}}$ with $T = B_T/B$ and $P = B_P/B$, where σ_B is the sign of B_ϕ and σ_I is the sign of I_ϕ. $B_T = \left|I\left(\psi\right)\right|/R$ and $B_P = \left\|\nabla\psi\right\|/R$ are definite positive and $\hat{\mathbf{s}} = \widehat{\phi}\times\widehat{\rho}$ with $\nabla\rho = \left\|\nabla\rho\right\|\widehat{\rho}$ such that $\hat{\mathbf{s}}$ is orientated counter-clockwise in the poloidal plane.

Expressing the scalar product $\mathbf{k}_\perp \cdot \check{\mathbf{k}}_\perp$ as

$$\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\cdot\left(\check{\mathbf{k}}\times\widehat{\mathbf{b}}\right) = \left(\mathbf{k}\cdot\check{\mathbf{k}}\right)\left(\widehat{\mathbf{b}}\cdot\widehat{\mathbf{b}}\right) - \left(\mathbf{k}\cdot\widehat{\mathbf{b}}\right)\left(\widehat{\mathbf{b}}\cdot\check{\mathbf{k}}\right) = \mathbf{k}\cdot\check{\mathbf{k}} - k_\parallel^2 \tag{5}$$

and using (4), (2) becomes

$$k_\perp\check{k}_\perp\cos\beta = k_\perp^2\chi\cos\beta = k_\rho\check{k}_\rho\left\|\nabla\rho\right\|^2 + \frac{m\check{m}}{r^2} + \frac{n\check{n}}{R^2} - \frac{\left\|\nabla\rho\right\|\sin\alpha}{r}\left(m\check{k}_\rho + \check{m}k_\rho\right) - k_\parallel^2 \tag{6}$$

since $\hat{\rho}\cdot\hat{\theta} = -\sin\alpha$ and $\hat{\theta}\cdot\hat{\phi} = \hat{\rho}\cdot\hat{\phi} = 0$ [5]. This the second equation between components of $\check{\mathbf{k}}$.

Since $\check{k}_\perp$ is determined from the dispersion relation from which $\check{k}_\perp$ is evaluated, the third equation is simply

$$\chi^2 k_\perp^2 = \check{k}_\rho^2\left\|\nabla\rho\right\|^2 - 2\check{k}_\rho\left\|\nabla\rho\right\|\frac{\check{m}\sin\alpha}{r} + \frac{\check{m}^2}{r^2} + \frac{\check{n}^2}{R^2} - k_\parallel^2 \tag{7}$$

The set of equations (3), (6) and (7) gives an implicit formulation of the components of $\check{\mathbf{k}}$ as a function of those of $\mathbf{k}$. However, even if $\check{k}_\rho$, $\check{m}$ and $\check{n}$ can be fully defined in this way, this formulation is not appropriate for fast numerical calculations, owing to the non-linearity of (7).

A more suitable approach for an efficient numerical implementation consists in deriving an explicit formulation of the components of $\check{\mathbf{k}}$ as a function of $\mathbf{k}$, β and χ, considering the geometrical transformation from $\mathbf{k}$ to $\check{\mathbf{k}}$. Let define the orthonormal frame of reference $\left(\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\times\widehat{\mathbf{b}}/k_\perp, \mathbf{k}\times\widehat{\mathbf{b}}/k_\perp, \widehat{\mathbf{b}}\right)$ formed by the local directions $\widehat{\mathbf{b}}$ and $\mathbf{k}$ of the magnetic field and wave vector respectively. The scattered wave vector may be expressed

$$\begin{aligned}\check{\mathbf{k}} &= \left(\check{\mathbf{k}}\cdot\widehat{\mathbf{b}}\right)\widehat{\mathbf{b}} + \left(\check{\mathbf{k}}\cdot\frac{\mathbf{k}\times\widehat{\mathbf{b}}}{k_\perp}\right)\frac{\mathbf{k}\times\widehat{\mathbf{b}}}{k_\perp} + \left(\check{\mathbf{k}}\cdot\left[\frac{\mathbf{k}\times\widehat{\mathbf{b}}}{k_\perp}\times\widehat{\mathbf{b}}\right]\right)\left[\frac{\mathbf{k}\times\widehat{\mathbf{b}}}{k_\perp}\times\widehat{\mathbf{b}}\right] \\ &= \check{k}_\parallel\widehat{\mathbf{b}} + \check{k}_\perp\sin\beta\frac{\mathbf{k}\times\widehat{\mathbf{b}}}{k_\perp} - \check{k}_\perp\cos\beta\frac{\mathbf{k}\times\widehat{\mathbf{b}}}{k_\perp}\times\widehat{\mathbf{b}} \\ &= \check{k}_\parallel\widehat{\mathbf{b}} + \chi\left(\sin\beta\mathbf{k}\times\widehat{\mathbf{b}} - \cos\beta\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\times\widehat{\mathbf{b}}\right)\end{aligned} \tag{8}$$

where

$$\check{k}_\perp = \check{\mathbf{k}} \cdot \frac{\mathbf{k} \times \widehat{\mathbf{b}}}{k_\perp} \tag{9}$$

and it can be easily verified that for $\beta = 0$ and $\chi = 1$, $\check{\mathbf{k}} = \mathbf{k}$ is well recovered, since $\check{k}_\parallel = k_\parallel$ in the model. From (8), the second right handside term corresponds to a rotation of an angle β combined with an homothetic transformation of the module of the perpendicular component of the wave vector.

The scattering operator $\mathcal{S}_\beta^\chi(\mathbf{k})$ is then

$$\mathcal{S}_\beta^\chi(\mathbf{k}) = \left(\mathbf{k} \cdot \widehat{\mathbf{b}}\right) \widehat{\mathbf{b}} + \chi \left(\sin\beta \mathbf{k} \times \widehat{\mathbf{b}} - \cos\beta \left(\mathbf{k} \times \widehat{\mathbf{b}}\right) \times \widehat{\mathbf{b}}\right) \tag{10}$$

and $\mathcal{S}_0^1 = \mathcal{I}$, where $\mathcal{I}$ is the identity transformation.

From (4),

$$\check{k}_\rho \|\nabla\rho\| - \frac{\check{m}}{r}\sin\alpha = k_\parallel \widehat{\mathbf{b}} \cdot \widehat{\rho} + \chi \sin\beta \mathbf{k} \times \widehat{\mathbf{b}} \cdot \widehat{\rho} - \chi\cos\beta \left(\mathbf{k} \times \widehat{\mathbf{b}}\right) \times \widehat{\mathbf{b}} \cdot \widehat{\rho} \tag{11}$$

$$-\check{k}_\rho \|\nabla\rho\| \sin\alpha + \frac{\check{m}}{r} = k_\parallel \widehat{\mathbf{b}} \cdot \widehat{\theta} + \chi \sin\beta \mathbf{k} \times \widehat{\mathbf{b}} \cdot \widehat{\theta} - \chi\cos\beta \left(\mathbf{k} \times \widehat{\mathbf{b}}\right) \times \widehat{\mathbf{b}} \cdot \widehat{\theta} \tag{12}$$

$$\frac{\check{n}}{R} = k_\parallel \widehat{\mathbf{b}} \cdot \widehat{\phi} + \chi \sin\beta \mathbf{k} \times \widehat{\mathbf{b}} \cdot \widehat{\phi} - \chi\cos\beta \left(\mathbf{k} \times \widehat{\mathbf{b}}\right) \times \widehat{\mathbf{b}} \cdot \widehat{\phi} \tag{13}$$

where all scalar products must be evaluated using the definitions of $\widehat{\rho}$, $\widehat{\theta}$ and $\widehat{\phi}$ introduced in Ref.[5], taking in particular account that $\hat{\rho} \cdot \hat{\theta} = -\sin\alpha$.

By combining (11) and (12),

$$\begin{aligned}\check{k}_\rho \|\nabla\rho\| \cos^2\alpha = {} & k_\parallel \widehat{\mathbf{b}} \cdot \widehat{\rho} + \chi \sin\beta \mathbf{k} \times \widehat{\mathbf{b}} \cdot \widehat{\rho} - \chi\cos\beta \left(\mathbf{k} \times \widehat{\mathbf{b}}\right) \times \widehat{\mathbf{b}} \cdot \widehat{\rho} \\ & + \left(k_\parallel \widehat{\mathbf{b}} \cdot \widehat{\theta} + \chi \sin\beta \mathbf{k} \times \widehat{\mathbf{b}} \cdot \widehat{\theta} - \chi\cos\beta \left(\mathbf{k} \times \widehat{\mathbf{b}}\right) \times \widehat{\mathbf{b}} \cdot \widehat{\theta}\right) \sin\alpha \end{aligned} \tag{14}$$

and

$$\begin{aligned}\frac{\check{m}}{r}\cos^2\alpha = {} & \left(k_\parallel \widehat{\mathbf{b}} \cdot \widehat{\rho} + \chi \sin\beta \mathbf{k} \times \widehat{\mathbf{b}} \cdot \widehat{\rho} - \chi\cos\beta \left(\mathbf{k} \times \widehat{\mathbf{b}}\right) \times \widehat{\mathbf{b}} \cdot \widehat{\rho}\right) \sin\alpha \\ & + k_\parallel \widehat{\mathbf{b}} \cdot \widehat{\theta} + \chi \sin\beta \mathbf{k} \times \widehat{\mathbf{b}} \cdot \widehat{\theta} - \chi\cos\beta \left(\mathbf{k} \times \widehat{\mathbf{b}}\right) \times \widehat{\mathbf{b}} \cdot \widehat{\theta} \end{aligned} \tag{15}$$

Evaluating $\left(\check{k}_\rho, \check{m}, \check{n}\right)$ requires to calculate all scalar products of the forms $\widehat{\mathbf{b}} \cdot \ldots$, $\mathbf{k} \times \widehat{\mathbf{b}} \cdot \ldots$ and $\left(\mathbf{k} \times \widehat{\mathbf{b}}\right) \times \widehat{\mathbf{b}} \cdot \ldots$ with $\ldots = \widehat{\rho}, \widehat{\theta}, \widehat{\phi}$.

$$\widehat{\mathbf{b}} \cdot \widehat{\rho} = 0 \tag{16}$$

$$\widehat{\mathbf{b}} \cdot \widehat{\theta} = \sigma_I P \cos\alpha \tag{17}$$

$$\widehat{\mathbf{b}} \cdot \widehat{\phi} = \sigma_B T \tag{18}$$

since $\widehat{\phi}\cdot\widehat{\rho}=\hat{\mathbf{s}}\cdot\widehat{\rho}=\widehat{\phi}\cdot\widehat{\theta}=\hat{\mathbf{s}}\cdot\widehat{\phi}=0$ and using the relation $\hat{\mathbf{s}}\cdot\hat{\theta}=\cos\alpha$. Using the vectorial transformations

$$\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\cdot\widehat{\rho}=\mathbf{k}\cdot\left(\widehat{\mathbf{b}}\times\widehat{\rho}\right)=\sigma_B T\mathbf{k}\cdot\hat{\mathbf{s}}-\sigma_I P\mathbf{k}\cdot\widehat{\phi} \tag{19}$$

$$\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\cdot\widehat{\theta}=\mathbf{k}\cdot\left(\widehat{\mathbf{b}}\times\widehat{\theta}\right)=-\sigma_B T\mathbf{k}\cdot\hat{\mathbf{r}}+\sigma_I P\sin\alpha\mathbf{k}\cdot\widehat{\phi} \tag{20}$$

$$\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\cdot\widehat{\phi}=\mathbf{k}\cdot\left(\widehat{\mathbf{b}}\times\widehat{\phi}\right)=\sigma_I P\mathbf{k}\cdot\widehat{\rho} \tag{21}$$

since

$$\begin{aligned}
\widehat{\mathbf{b}}\times\widehat{\rho}&=\sigma_B T\widehat{\phi}\times\widehat{\rho}+\sigma_I P\hat{\mathbf{s}}\times\widehat{\rho}=\sigma_B T\hat{\mathbf{s}}-\sigma_I P\widehat{\phi}\\
\widehat{\mathbf{b}}\times\widehat{\theta}&=\sigma_B T\widehat{\phi}\times\widehat{\theta}+\sigma_I P\hat{\mathbf{s}}\times\widehat{\theta}=-\sigma_B T\hat{\mathbf{r}}+\sigma_I P\sin\alpha\widehat{\phi}\\
\widehat{\mathbf{b}}\times\widehat{\phi}&=\sigma_B T\widehat{\phi}\times\widehat{\phi}+\sigma_I P\hat{\mathbf{s}}\times\widehat{\phi}=\sigma_I P\widehat{\rho}
\end{aligned}$$

and

$$\begin{aligned}
\widehat{\phi}\times\widehat{\rho}&=\hat{\mathbf{s}}\\
\widehat{\phi}\times\widehat{\theta}&=-\hat{\mathbf{r}}\\
\hat{\mathbf{s}}\times\widehat{\rho}&=-\widehat{\phi}\\
\hat{\mathbf{s}}\times\widehat{\theta}&=\sin\alpha\widehat{\phi}\\
\hat{\mathbf{s}}\times\widehat{\phi}&=\widehat{\rho}
\end{aligned}$$

where $\hat{\mathbf{s}}=\sin\alpha\widehat{\mathbf{r}}+\cos\alpha\widehat{\theta}$.

Finally, since

$$\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\times\widehat{\mathbf{b}}\cdot\widehat{\rho}=k_{\parallel}\widehat{\mathbf{b}}\cdot\widehat{\rho}-\mathbf{k}\cdot\widehat{\rho}=-\mathbf{k}\cdot\widehat{\rho} \tag{22}$$

$$\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\times\widehat{\mathbf{b}}\cdot\widehat{\theta}=k_{\parallel}\widehat{\mathbf{b}}\cdot\widehat{\theta}-\mathbf{k}\cdot\widehat{\theta}=k_{\parallel}\sigma_I P\cos\alpha-\mathbf{k}\cdot\widehat{\theta} \tag{23}$$

$$\left(\mathbf{k}\times\widehat{\mathbf{b}}\right)\times\widehat{\mathbf{b}}\cdot\widehat{\phi}=k_{\parallel}\widehat{\mathbf{b}}\cdot\widehat{\phi}-\mathbf{k}\cdot\widehat{\phi}=k_{\parallel}\sigma_B T-\mathbf{k}\cdot\widehat{\phi} \tag{24}$$

and

$$\begin{aligned}
\mathbf{k}\cdot\hat{\mathbf{s}}&=\frac{m}{r}\cos\alpha\\
\mathbf{k}\cdot\widehat{\phi}&=\frac{n}{R}\\
\mathbf{k}\cdot\widehat{\mathbf{r}}&=k_\rho\left\|\nabla\rho\right\|\cos\alpha\\
\mathbf{k}\cdot\widehat{\rho}&=k_\rho\left\|\nabla\rho\right\|-\sin\alpha\frac{m}{r}\\
\mathbf{k}\cdot\widehat{\theta}&=-\sin\alpha k_\rho\left\|\nabla\rho\right\|+\frac{m}{r}
\end{aligned}$$

using the fact that $\hat{\rho} \cdot \hat{\mathbf{r}} = \cos\alpha$, the components of $\check{\mathbf{k}}$ are

$$\begin{aligned}\check{k}_\rho \left\|\nabla\rho\right\| \cos^2\alpha = \chi \sin\beta \left(\sigma_B T \frac{m}{r} \cos\alpha - \sigma_I P \frac{n}{R}\right) \\ + \chi \cos\beta k_\rho \left\|\nabla\rho\right\| + k_\parallel \sigma_I P \cos\alpha \sin\alpha \\ + \chi \sin\beta \left(-\sigma_B T k_\rho \left\|\nabla\rho\right\| \cos\alpha + \sigma_I P \sin\alpha \frac{n}{R}\right) \sin\alpha \\ - \chi \cos\beta \left(k_\parallel \sigma_I P \cos\alpha + \sin\alpha k_\rho \left\|\nabla\rho\right\|\right) \sin\alpha \end{aligned} \quad (25)$$

$$\begin{aligned}\frac{\check{m}}{r} \cos\alpha = +k_\parallel \sigma_I P \\ + \chi \sin\beta \sin\alpha \sigma_B T \frac{m}{r} + \chi \cos\beta \cos\alpha \frac{m}{r} \\ - \chi \sin\beta \sigma_B T k_\rho \left\|\nabla\rho\right\| - \chi \cos\beta k_\parallel \sigma_I P \end{aligned} \quad (26)$$

$$\frac{\check{n}}{R} = k_\parallel \sigma_B T + \chi \sigma_I P \sin\beta \left(k_\rho \left\|\nabla\rho\right\| - \sin\alpha \frac{m}{r}\right) - \chi \cos\beta \left(k_\parallel \sigma_B T - \frac{n}{R}\right) \quad (27)$$

By combining (26) and (27), it can be shown, that relation (3) is well satisfied. In Appendix A, it is also demonstrated that using relations (25), (26) and (27), both relations (6) and (7) are also satisfied, which demonstrate the consistency between the geometrical (explicit) and the physical (implicit) formulation of $\check{\mathbf{k}}$.

2 The wave kinetic equation

3 Numerical algorithm

4 Conclusions

References

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A Appendix A

The two first components of $\check{\mathbf{k}}$ may be expressed in a slightly modified form as compared to (25) and (26),

$$\check{k}_\rho\|\nabla\rho\| = \sigma_I P\tan\alpha k_\parallel + \chi\sin\beta\left[\frac{\sigma_B T}{\cos\alpha}\left(\frac{m}{r} - k_\rho\|\nabla\rho\|\sin\alpha\right) - \sigma_I P\frac{n}{R}\right] -$$

$$-\chi\cos\beta\left(\sigma_I P\tan\alpha k_\parallel - k_\rho\|\nabla\rho\|\right) \tag{28}$$

$$\frac{\check{m}}{r} = \frac{\sigma_I P}{\cos\alpha}k_\parallel + \chi\sin\beta\frac{\sigma_B T}{\cos\alpha}\left(\frac{m}{r}\sin\alpha - k_\rho\|\nabla\rho\|\right) - \chi\cos\beta\left(\frac{\sigma_I P}{\cos\alpha}k_\parallel - \frac{m}{r}\right) \tag{29}$$

which is more convenient for verifying that the geometrical approach based on the scattering operator $\mathcal{S}_\beta^\chi$ defined by relation (10) is well consistent with the physics of the scattering. We have to check that these transformed components satisfy the conditions (1), (2) and

$$\check{k}_\perp^2 = \check{k}_\rho^2\|\nabla\rho\|^2 - 2\check{k_\rho}\|\nabla\rho\|\frac{\check{m}\sin\alpha}{r} + \frac{\check{m}^2}{r^2} + \frac{\check{n}^2}{R^2} - k_\parallel^2 \tag{30}$$

Using the components of the wave vector $\mathbf{k}$ and$\check{\mathbf{k}}$ before and after the scattering event, these conditions can be written in the form:

$$\sigma_I P\frac{\check{m}}{r}\cos\alpha + \sigma_B T\frac{\check{n}}{R} = \sigma_I P\frac{m}{r}\cos\alpha + \sigma_B T\frac{n}{R} \tag{31}$$

$$k_\rho\check{k}_\rho\|\nabla\rho\|^2 + \frac{m\check{m}}{r^2} + \frac{n\check{n}}{R^2} - \frac{\|\nabla\rho\|\sin\alpha}{r}\left(m\check{k}_\rho + \check{m}k_\rho\right) = k_\perp^2\chi\cos\beta \tag{32}$$

$$\check{k}_\rho^2\|\nabla\rho\|^2 - 2\check{k}_\rho\|\nabla\rho\|\frac{\check{m}\sin\alpha}{r} + \frac{\check{m}^2}{r^2} + \frac{\check{n}^2}{R^2} - k_\parallel^2 = \chi^2 k_\perp^2 \tag{33}$$

A.1 Conservation of $k_\parallel$

$$\check{k}_\parallel = \sigma_I P \frac{\check{m}}{r}\cos\alpha + \sigma_B T \frac{\check{n}}{R} =$$

$$= \sigma_I P \cos\alpha \left[\frac{\sigma_I P}{\cos\alpha}k_\parallel + \chi \sin\beta \frac{\sigma_B T}{\cos\alpha}\left(\frac{m}{r}\sin\alpha - k_\rho\|\nabla\rho\|\right) - \chi\cos\beta\left(\frac{\sigma_I P}{\cos\alpha}k_\parallel - \frac{m}{r}\right)\right] +$$

$$+\sigma_B T\left[\sigma_B T k_\parallel + \chi\sin\beta\sigma_I P\left(k_\rho\|\nabla\rho\| - \frac{m}{r}\sin\alpha\right) - \chi\cos\beta\left(\sigma_B T k_\parallel - \frac{n}{R}\right)\right] =$$

$$= k_\parallel\left(\sigma_I^2 P^2 + \sigma_B^2 T^2\right) + \sigma_I P \sigma_B T \chi \sin\beta\left(\frac{m}{r}\sin\alpha - k_\rho\|\nabla\rho\| + k_\rho\|\nabla\rho\| - \frac{m}{r}\sin\alpha\right) -$$

$$-\chi\cos\beta\left[k_\parallel\left(\sigma_I^2 P^2 + \sigma_B^2 T^2\right) - \left(\sigma_I P\frac{m}{r}\cos\alpha + \sigma_B T\frac{n}{R}\right)\right] =$$

$$= k_\parallel - \chi\cos\beta\left(k_\parallel - k_\parallel\right) = k_\parallel$$

A.2 Direction of $\check{\mathbf{k}}_\perp$

$$\mathbf{k}_\perp \cdot \check{\mathbf{k}}_\perp = k_\rho\check{k}_\rho\|\nabla\rho\|^2 + \frac{m\check{m}}{r^2} + \frac{n\check{n}}{R^2} - \frac{\|\nabla\rho\|\sin\alpha}{r}\left(m\check{k}_\rho + \check{m}k_\rho\right) - k_\parallel^2 =$$

$$= k_\rho\|\nabla\rho\|\left\{\sigma_I P\tan\alpha k_\parallel + \chi\sin\beta\left[\frac{\sigma_B T}{\cos\alpha}\left(\frac{m}{r} - k_\rho\|\nabla\rho\|\sin\alpha\right) - \sigma_I P\frac{n}{R}\right]\right\} -$$

$$-k_\rho\|\nabla\rho\|\chi\cos\beta\left(\sigma_I P\tan\alpha k_\parallel - k_\rho\|\nabla\rho\|\right) +$$

$$+\frac{m}{r}\left[\frac{\sigma_I P}{\cos\alpha}k_\parallel + \chi\sin\beta\frac{\sigma_B T}{\cos\alpha}\left(\frac{m}{r}\sin\alpha - k_\rho\|\nabla\rho\|\right) - \chi\cos\beta\left(\frac{\sigma_I P}{\cos\alpha}k_\parallel - \frac{m}{r}\right)\right] +$$

$$+\frac{n}{R}\left[\sigma_B T k_\parallel + \chi\sin\beta\sigma_I P\left(k_\rho\|\nabla\rho\| - \frac{m}{r}\sin\alpha\right) - \chi\cos\beta\left(\sigma_B T k_\parallel - \frac{n}{R}\right)\right] -$$

$$-\frac{m}{r}\sin\alpha\left\{\sigma_I P\tan\alpha k_\parallel + \chi\sin\beta\left[\frac{\sigma_B T}{\cos\alpha}\left(\frac{m}{r} - k_\rho\|\nabla\rho\|\sin\alpha\right) - \sigma_I P\frac{n}{R}\right]\right\} +$$

$$+\frac{m}{r}\sin\alpha\chi\cos\beta\left(\sigma_I P\tan\alpha k_\parallel - k_\rho\|\nabla\rho\|\right) -$$

$$-k_\rho\|\nabla\rho\|\sin\alpha\left[\frac{\sigma_I P}{\cos\alpha}k_\parallel + \chi\sin\beta\frac{\sigma_B T}{\cos\alpha}\left(\frac{m}{r}\sin\alpha - k_\rho\|\nabla\rho\|\right) - \chi\cos\beta\left(\frac{\sigma_I P}{\cos\alpha}k_\parallel - \frac{m}{r}\right)\right] - k_\parallel^2 =$$

$$= \left[\left(\sigma_I P \frac{m}{r}\cos\alpha + \sigma_B T \frac{n}{R}\right) k_\parallel - k_\parallel^2\right] + \chi\cos\beta\left[k_\rho^2\|\nabla\rho\|^2 + \frac{m^2}{r^2} + \frac{n^2}{R^2} - 2k_\rho\|\nabla\rho\|\sin\alpha\right] -$$

$$-\chi\cos\beta k_\parallel\left(\sigma_I P\frac{m}{r}\cos\alpha + \sigma T\frac{n}{R}\right) =$$

$$= \left(k_\parallel^2 - k_\parallel^2\right) + \chi\cos\beta\left(k^2 - k_\parallel^2\right) = k_\perp^2\chi\cos\beta$$

A.3 Solution of the dispersion relation

$$\check{k}_\perp^2 = \check{k}_\rho^2\|\nabla\rho\|^2 - 2\check{k}_\rho\|\nabla\rho\|\frac{\check{m}\sin\alpha}{r} + \frac{\check{m}^2}{r^2} + \frac{\check{n}^2}{R^2} - k_\parallel^2 =$$

$$= \left\{\sigma_I P\tan\alpha k_\parallel + \chi\sin\beta\left[\frac{\sigma_B T}{\cos\alpha}\left(\frac{m}{r} - k_\rho\|\nabla\rho\|\sin\alpha\right) - \sigma_I P\frac{n}{R}\right] - \chi\cos\beta\left(\sigma_I P\tan\alpha k_\parallel - k_\rho\|\nabla\rho\|\right)\right\}^2 -$$

$$-2\sin\alpha\left[\frac{\sigma_I P}{\cos\alpha}k_\parallel + \chi\sin\beta\frac{\sigma_B T}{\cos\alpha}\left(\frac{m}{r}\sin\alpha - k_\rho\|\nabla\rho\|\right) - \chi\cos\beta\left(\frac{\sigma_B T}{\cos\alpha}k_\parallel - \frac{m}{r}\right)\right]\cdot$$

$$\cdot\left\{\sigma_I P\tan\alpha k_\parallel + \chi\sin\beta\left[\frac{\sigma_B T}{\cos\alpha}\left(\frac{m}{r} - k_\rho\|\nabla\rho\|\sin\alpha\right) - \sigma_I P\frac{n}{R}\right] - \chi\cos\beta\left(\sigma_I P\tan\alpha k_\parallel - k_\rho\|\nabla\rho\|\right)\right\} +$$

$$+\left[\frac{\sigma_I P}{\cos\alpha}k_\parallel + \chi\sin\beta\frac{\sigma_B T}{\cos\alpha}\left(\frac{m}{r}\sin\alpha - k_\rho\|\nabla\rho\|\right) - \chi\cos\beta\left(\frac{\sigma_I P}{\cos\alpha} - \frac{m}{r}\right)\right]^2 +$$

$$+\left[\sigma_B T k_\parallel + \chi\sin\beta\sigma_I P\left(k_\rho\|\nabla\rho\| - \frac{m}{r}\sin\alpha\right) - \chi\cos\beta\left(\sigma_B T k_\parallel - \frac{n}{R}\right)\right]^2 - k_\parallel^2 =$$

$$= \left[\left(\sigma_I^2 P^2 + \sigma_B^2 T^2\right)k_\parallel^2 - k_\parallel^2\right] + 2\chi\sin\beta\sigma_I P\sigma_B T k_\parallel\left(k_\rho\|\nabla\rho\| + \frac{m}{r}\sin\alpha - k_\rho\|\nabla\rho\| - \frac{m}{r}\sin\alpha\right) -$$

$$-2\chi\cos\beta\left[\left(\sigma_I^2 P^2 + \sigma_B^2 T^2\right)k_\parallel^2 - \left(\sigma_I P\frac{m}{r}\cos\alpha + \sigma_B T\frac{n}{R}\right)k_\parallel\right] -$$

$$-2\chi^2\sin\beta\cos\beta\left(\frac{m}{r}\sin\alpha - k_\rho\|\nabla\rho\|\right)\left[\frac{\sigma_B T}{\cos\alpha}\left(\frac{m}{r} - k_\rho\|\nabla\rho\|\sin\alpha\right) + \sigma_I P\left(\sigma_B T k_\parallel - \frac{n}{R}\right)\right] +$$

$$+2\chi^2\sin\beta\cos\beta\left(\frac{m}{r}\sin\alpha - k_\rho\|\nabla\rho\|\right)\left[\frac{\sigma_B T}{\cos\alpha}\left(\frac{m}{r} - k_\rho\|\nabla\rho\|\sin\alpha\right) + \sigma_I P\left(\sigma_B T k_\parallel - \frac{n}{R}\right)\right] +$$

$$+\chi^2 \sin^2\beta \left[\left(k_\rho \|\nabla\rho\| - \frac{m}{r}\sin\alpha \right)^2 + \left(\sigma_B T \frac{n}{R}\cos\alpha - \sigma_I P \frac{n}{R} \right)^2 \right] +$$

$$+\chi^2 \cos^2\beta \left(k_\rho^2 \|\nabla\rho\|^2 - 2\frac{m}{r} k_\rho \|\nabla\rho\| \sin\alpha + \frac{m^2}{r^2} + \frac{n^2}{R^2} - k_\parallel^2 \right) =$$

$$= \left(k_\parallel^2 - k_\parallel^2 \right) - 2\chi\cos\beta \left(k_\parallel^2 - k_\parallel^2 \right) + \chi^2 k_\perp^2 \sin^2\beta + \chi^2 \cos^2\beta \left(k^2 - k_\parallel^2 \right) =$$

$$= \chi^2 k_\perp^2 \left(\sin^2\beta + \cos^2\beta \right) = \chi^2 k_\perp^2$$

The expressions for the component of the scattered wave vector derived using the scattering operator satisfy the conservation of $k_\parallel$, the relation between $\mathbf{k}_\perp$and $\check{\mathbf{k}}_\perp$and the dispersion relation.

B Appendix B

Depending upon the plasma regime and the spatial localization, magnetic or density fluctuation spectra may be well approximated by Gaussian or Cauchy-Lorentz distributions. For this purpose, appropriate random numbers must be generated with the corresponding probability distribution function (pdf). Let denotes the uniform pdf for a variable x as

$$p(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & otherwise \end{cases} \tag{34}$$

and a non-uniform pdf for the variable y denoted $p(y)\,dy$ which related to $p(x)\,dx$ by the fundamental transformation law of probabilities

$$|p(y)\,dy| = |p(x)\,dx| \tag{35}$$

so that

$$p(y) = \left| \frac{dx}{dy} \right| p(x) \tag{36}$$

If the relation $y(x)$ is explicitely known, then it is possible to generate a random numbers with a pdf $p(y)$ from a the uniform one $p(x)$ by applying the transformation

$$y = F^{-1}(x) \tag{37}$$

where $F^{-1}(y)$ is the inverse of the indefinite integral of the function $f(y) = dx/dy$. Wether (37) is feasible to implement depends on wether the inverse function of the integral of $f(y)$is itself feasible to compute either analytically or numerically. Here F is nothing but the cumulative distribution function (cdf) associated to $p(y)$.

B.1 Cauchy-Lorentz distribution

For the Cauchy-Lorentz distribution centered around $y = 0$, the cdf takes the simple form

$$F(y) = \frac{1}{2} + \frac{1}{\pi} \tan^{-1}\left(\frac{2y}{\Delta}\right) \tag{38}$$

where Δ is the full width at half maximum (FWHM). It's inverse value is then

$$F^{-1}(x) = \frac{\Delta}{2} \tan\left(\pi\left(x - \frac{1}{2}\right)\right) \tag{39}$$

and the calculation of random numbers with a Cauchy-Lorentz pdf from a uniform random generator is straightforward.

The resulting distribution is

$$p(y) = \frac{1}{\pi} \frac{\Delta/2}{y^2 + (\Delta/2)^2} \tag{40}$$

B.2 Gaussian distribution

For a one dimensional Gaussian or normal distribution,

$$p(y) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{y^2}{2\sigma^2}\right) \tag{41}$$

with $y \in]-\infty, +\infty[$, the approach is more complex, since F^{-1} does not exist analytically. In (41), the variance σ is related to the full width at half maximum by the usual relation $\Delta = 2\sqrt{2\ln 2}\sigma$.

Instead of considering the usual Box-Muller method for a two dimensional problem [6], an efficient standard rejection technique is applied in a first step to the normalized Gaussian distribution $p^+(y)$ on the restricted domain $y \in [0, +\infty[$

$$p^+(y) = \frac{1}{\sigma}\sqrt{\frac{2}{\pi}} \exp\left(-\frac{y^2}{2\sigma^2}\right) \tag{42}$$

Since the pdf (42) may be expressed in the form

$$p^+(y) = \frac{1}{\sigma}\sqrt{\frac{2}{\pi}} \exp\left(\frac{1}{2\sigma^2}\right) \exp\left(-\frac{y}{\sigma^2}\right) \exp\left(-\frac{(y-1)^2}{2\sigma^2}\right) \tag{43}$$

it is possible to apply the rejection technique but for random numbers which are non uniformly distributed. Indeed, the pdf (43) may be interpreted as a two dimensional probability distribution

$$\widetilde{p}^+(y, z) \equiv \exp\left(-\frac{y}{\sigma^2}\right) \exp\left(-\frac{z}{\sigma^2}\right) \tag{44}$$

with the usual constraint of acceptance for the coordinates (y, z) [6],

$$-\frac{z}{\sigma^2} < -\frac{(y-1)^2}{2\sigma^2} \tag{45}$$

or

$$\frac{(y-1)^2}{2} < z \tag{46}$$

It is important to note that the normalisation factors in front of (43) do not play a role in the shape of the distribution. So they do not contribute to the generation of random numbers y with a Gaussian distribution.

The rejection procedure starts with the generation of two random numbers $y[z]$ whose pdf has the same form $\exp\left(-y[z]/\sigma^2\right)$in the interval $[y[z], y[z] + dy[dz]]$, according to (44). This is achieved by the relation

$$y[z] = -\sigma^2 \ln(x_y[x_z]) \tag{47}$$

where $x_y[x_z] \in]0,1[$ [6]. Once $y[z]$ have been obtained, the value y is accepted if the condition (46) is fullfiled. Otherwise the whole procedure is repeated until the value y is accepted.

The probability of obtaining a satisfactory random number y on the first trial, which called the efficiency $\mathcal{E}$ of the random generator, is the sum of the probabilities of selecting a value y in any one of the separate interval dy,

$$\begin{aligned}\mathcal{E} &= \int_0^{+\infty} \exp\left(-\frac{y}{\sigma^2}\right) \exp\left(-\frac{(y-1)^2}{2\sigma^2}\right) dy \\ &= \sigma\sqrt{\frac{\pi}{2}} \exp\left(-\frac{1}{2\sigma^2}\right)\end{aligned} \tag{48}$$

For $\sigma = 1$, $\mathcal{E} \simeq 0.76$, which is a high number for a rejection method. However, as shown in Fig.1, the performances of the Gaussian random generator degradate rapidly when the variance σ is less than unity, otherwise, $\mathcal{E}$ scales almost linearly with σ.

Finally, in order to get $p(y)$ in the interval $y \in]-\infty, +\infty[$, the solution y obtained with (47) must be multiplied by the random number $s = sign(2x_s - 1)$, where x_s is another uniform random number itself in the interval $x_s \in]0,1[$.

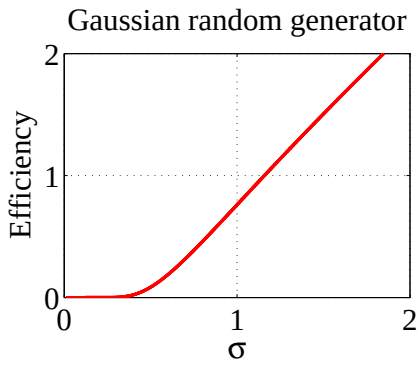


Figure 1: Efficiency of the Gaussian random number generator as a function of the variance